CHAPTER 17

Simple Interest

INTRODUCTION

When a person A borrows some money from another person B, then A has to pay certain amount to B in exchange of using the money. This amount paid by A is called *Interest*. The total amount of money borrowed by A from B is called the *Principal*. The money paid back to B, which comprises the principal and the interest is called the *Amount.*

In other words,

Amount = Principal + Interest

The interest is usually charged according to a specified term, which is expressed as some per cent of the principal and is called the *rate of interest* for a fixed period of time. This fixed period may be a year, six months, three months or a month, and correspondingly the rate of interest is charged annually, semi-annually, quarterly or monthly basis. For Examinationple, the rate of interest is 5% per annum means, the interest payable on ₹100 for one year is ₹5.

Interest is of two types:

1. Simple Interest
2. Compound Interest

SIMPLE INTEREST

When interest is payable on the principal amount only, it is called *Simple interest*. For example, simple interest on ₹100 at 5% per annum will be ₹5 each year, that is, at the end of the first year, total amount will be ₹105. At the end of the second year, it will be ₹110, and so on.

Thus, simple interest is the interest computed on the principal amount for the entire period it is borrowed.

In this chapter, we shall limit ourselves to simple interest. Compound interest will be discussed in the next chapter.

BASIC FORMULAE

If P stands for principal, R is the rate per cent per annum, T is the number of years, I is the simple interest and A is the amount, then

01 $$\text{Simple Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

or, $$I = \frac{P \times R \times T}{100}$$

Illustration 1: Find the simple interest on ₹5200 for 2 years at 6% per annum.

Solution: Here, P = ₹5200, T = 2 years and R = 6%

$$\therefore \quad \text{Simple interest} = \frac{P \times R \times T}{100} = \frac{5200 \times 6 \times 2}{100}$$

= ₹624.

02 $$\text{Principal} = \frac{100 \times \text{Simple Interest}}{\text{Rate} \times \text{Time}}$$

or, $$P = \frac{100 \times I}{R \times T}$$

Illustration 2: A man earns ₹450 as interest in 2 years on a certain sum of money invested in a company at the rate of 12 per cent per annum. Find the sum invested by the man in the company.

Solution: We have, I = ₹450, T = 2 years,

R = 12% per annum

$$\therefore \quad P = \frac{I \times 100}{R \times T} = \frac{450 \times 100}{12 \times 2} = ₹1875.$$

Thus, the money invested by the man was ₹1875.

03 $\text{Rate} = \dfrac{100 \times \text{Simple Interest}}{\text{Principal} \times \text{Time}}$

or $R = \dfrac{100 \times I}{P \times T}$

Illustration 3: At what interest rate per annum, in 4 years, a sum of ₹5000 will become ₹6000?

Solution: Here, P = ₹5000, A = ₹6000, T = 4 years

So, $I = A - P$ = ₹(6000 − 5000) = ₹1000

$\therefore \quad R = \dfrac{100 \times I}{P \times T} = \dfrac{100 \times 1000}{5000 \times 4} = 5\%$

04 $\text{Time} = \dfrac{100 \times \text{Simple Interest}}{\text{Rate} \times \text{Principal}}$

or, $T = \dfrac{100 \times I}{R \times P}$

Illustration 4: In what time ₹1200 will earn an interest of ₹240 at 5% per annum?

Solution: Here, P = ₹1200, I = ₹240, R = 5%

$\therefore \quad T = \dfrac{100 \times I}{R \times P} = \dfrac{100 \times 240}{1200 \times 5} = 4$ years.

05 Amount = Principal + Simple Interest

$= \text{Principal} + \dfrac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$= \text{Principal}\left(1 + \dfrac{\text{Rate} \times \text{Time}}{100}\right)$

or, $A = P\left(1 + \dfrac{R \times T}{100}\right)$

Illustration 5: Mahesh borrowed ₹3000 from his friend Suresh at 15 per cent per annum for 3 years. Find the interest and money returned by Mahesh to Suresh.

Solution: Here, P = ₹3000, R = 15% per annum, T = 3 years

$\therefore \quad I = \dfrac{P \times R \times T}{100} = \dfrac{3000 \times 15 \times 3}{100} = ₹1350.$

$\therefore \quad A = P + I$ = ₹3000 + ₹1350 = ₹4350.

Thus, Mahesh paid ₹1350 as interest to Suresh, the total amount Mahesh returned to Suresh = ₹4350.

SHORT-CUT METHODS

01 If a certain sum in T years at R% per annum amounts to ₹A, then the sum will be

$$P = \frac{100 \times A}{100 + R \times T}$$

Explanation

Let the principal be ₹x

$\therefore$ Simple interest = ₹$(A - x)$

$\therefore \quad A - x = \dfrac{x \times R \times T}{100}$

$\Rightarrow \quad 100A - 100x = xRT$

$\Rightarrow \quad (100 + RT)x = 100A$

$\therefore \quad x = \dfrac{100 \times A}{100 + R \times T}.$

Illustration 6: What principal will amount to ₹570 at 4% per annum in $3\frac{1}{2}$ years?

Solution: We have, A = ₹570, R = 4% per annum, $T = \dfrac{7}{2}$ years.

$\therefore \quad P = \dfrac{100 \times A}{100 + R \times T} = \dfrac{100 \times 570}{100 + 4 \times 7/2}$

$= \dfrac{100 \times 570}{114} = ₹500$

Thus, ₹500 will be ₹570 at 4% per annum in $3\frac{1}{2}$ years.

02 The annual payment that will discharge a debt of ₹A due in T years at R% per annum is

$$\text{Annual payment} = ₹\left(\frac{100A}{100T + \dfrac{RT(T-1)}{2}}\right)$$

Explanation

Let the annual payment be ₹x.

Since the first instalment is paid at the end of the first year,

$\therefore$ Amount of the first instalment at the end of t years

$$= x + \frac{(T-1) \times R \times x}{100}$$

Similarly, amount of the second instalment at the end of t years

$$= x + \frac{(T-2)\times R\times x}{100}, \text{ and so on.}$$

Thus, total amount of T instalments

$$A = \left[x + \frac{(T-1)\times R\times x}{100}\right] + \left[x + \frac{(T-2)\times R\times x}{100}\right] + \cdots + x$$

$$= Tx + \frac{Rx}{100}[(T-1)+(T-2)+\cdots+1]$$

$$= Tx + \frac{Rx}{100}\left[\frac{(T-1)\times T}{2}\right]$$

or, $100\ Tx + Rx\left[\frac{T(T-1)}{2}\right] = 100A$

or, $x\left[100T + \frac{RT(T-1)}{2}\right] = 100A$

$$\therefore\quad x = \frac{100A}{100T + \frac{RT(T-1)}{2}}.$$

Illustration 7: Find out the annual instalment that will discharge a debt of ₹12,900 due in 4 years at 5% per annum simple interest.

Solution: Here, A = ₹12900, T = 4 years, R = 5% per annum.

$$\therefore\quad \text{Annual instalment} = \frac{100\times A}{100\times T + \frac{RT(T-1)}{2}}$$

$$= \frac{100\times 12900}{(100\times 4) + \frac{5(4-1)\times 4}{2}}$$

$$= \frac{100\times 12900}{400+30} = \frac{100\times 12900}{430}$$

$$= ₹3000.$$

03 If a certain sum is invested in n types of investments in such a manner that equal amount is obtained on each investment where interest rates are R_1, R_2, R_3, ..., R_n respectively, and time periods are T_1, T_2, T_3, ...,T_n respectively, then the ratio in which the amounts are invested is:

$$\frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} :$$

$$\frac{1}{100+R_3T_3} : \ldots \frac{1}{100+R_nT_n}.$$

Explanation

Let P_1, P_2, ..., P_n be invested in n types of investments whose interest rates are R_1, R_2, ..., R_n and time periods are T_1, T_2, ..., T_n.

Then, $P_1 = \frac{100\times A}{100+R_1T_1}$

$P_2 = \frac{100\times A}{100+R_2T_2}$

$\vdots \qquad \vdots$

$P_n = \left(\frac{100\times A}{100+R_nT_n}\right).$

$\therefore\quad P_1:P_2:\ldots:P_n$

$$= \frac{100\times A}{100+R_1T_1} : \frac{100\times A}{100+R_2T_2} : \ldots \frac{100\times A}{100+R_nT_n}$$

$$= \frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} : \ldots \frac{1}{100+R_nT_n}$$

[$\because$ the amount A remains same for all]

Illustration 8: A sum of ₹1586 is divided among three such parts that amount obtained on these three parts of money after 2, 3 and 4 years, respectively, at the rate of 5% per annum remains equal. Find out such three parts of the sum.

Solution: Since the amount accured from each of the three parts of ₹1586 at the rate of 5% p.a. in 2, 3 and 4 years, respectively, remains equal, such three parts of ₹1586 will be in the ratio of

$$\frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} : \frac{1}{100+R_3T_3}$$

Hence, the ratio is

$$= \frac{1}{100+5\times 2} : \frac{1}{100+5\times 3} : \frac{1}{100+5\times 4}$$

$$= \frac{1}{110} : \frac{1}{115} : \frac{1}{120}$$

$$= \frac{1\times 30360}{110} : \frac{1\times 30360}{115} : \frac{1\times 30360}{120}$$

($\because$ L.C.M. of 110, 115 and 120 is 30360)

$\therefore$ Ratio = 276:264:253

Sum of proportionals = 276 + 264 + 253 = 793

$\therefore$ 1st part = $\frac{276}{793}\times 1586$ = ₹552,

2nd part = $\frac{264}{793}\times 1586$ = ₹528

and, 3rd part = $\frac{253}{793}\times 1586$ = ₹506.

04 If a certain sum of money becomes n times itself in T years at simple interest, then the rate of interest per annum is

$$R = \frac{100(n-1)}{T}\%$$

Explanation

Let, ₹P become ₹nP in t years.

$\therefore$ Simple interest I is given by

$$I = P - P = (n-1)P$$

$\therefore$ Rate of interest R is given by

$$R = \frac{100\times I}{P\times T} = \frac{100\times(n-1)P}{P\times T} = \frac{100(n-1)}{T}.$$

Illustration 9: A certain sum of money trebles itself in 5 years simple interest. Find the rate per cent per annum.

Solution: Here, $n = 3$, $T = 5$ years

$$\therefore \quad R = \frac{100(n-1)}{T}\% = \frac{100(3-1)}{5}\% = 40\%$$

05 If a certain sum of money becomes n times itself at R% per annum simple interest in T years, then

$$T = \left(\frac{n-1}{R}\right) \times 100 \text{ years.}$$

Illustration 10: In what time a sum of money will double itself at a rate of simple interest of 8% p.a.?

Solution: Required time $(T) = \dfrac{(n-1)\times 100}{R}$ years

$$= \frac{(2-1)\times 100}{8} \text{ years}$$

$$= 12\frac{1}{2} \text{ years.}$$

06 If a certain sum of money becomes n times itself in T years at a simple interest, then the time T in which it will become m times itself is given by

$$T' = \left(\frac{m-1}{n-1}\right)\times T \text{ years.}$$

Explanation

Let the principal be ₹P.

Let it become m times in T' years.

Then, the amount in T years = ₹nP

and the amount in T' years = ₹mP.

$$\therefore \quad nP - P = \frac{P\times R\times T}{100}$$

$$\text{or,} \quad (n-1)P = \frac{P\times R\times T}{100} \qquad \cdots(1)$$

$$\text{and,} \quad (m-1)P = \frac{P\times R\times T}{100} \qquad \cdots(2)$$

$$\therefore \quad \frac{(m-1)P}{(n-1)P} = \frac{P\times R\times T'}{100}\times\frac{100}{P\times R\times T}$$

$$\text{or,} \quad \frac{m-1}{n-1} = \frac{T'}{T}$$

$$\therefore \quad T' = \left(\frac{m-1}{n-1}\right) T \text{ years.}$$

Illustration 11: A sum of money put out on simple interest doubles itself in $12\frac{1}{2}$ years. In how many years would it treble itself?

Solution: Here, $n = 2$, $m = 3$, $T = \dfrac{25}{2}$ years.

$$\therefore \quad \text{Required time } (T') = \left(\frac{m-1}{n-1}\right) \times T \text{ years}$$

$$= \left(\frac{3-1}{2-1}\right)\times\frac{25}{2} \text{ years}$$

$$= 25 \text{ years.}$$

07 Effect of change of P, R and T on simple interest is given by the following formula:

Change in Simple Interest

$$= \frac{\text{Product of fixed parameter}}{100}$$

$\times$ [difference of product of variable parameters]

For example, if rate (R) changes from R_1 to R_2 and P, T are fixed, then

$$\text{Change in SI} = \frac{PT}{100}\times(R_1 - R_2)$$

Similarly, if principal (P) changes from P_1 to P_2 and R, T are fixed, then change in SI $= \dfrac{RT}{100}\times(P_1 - P_2)$

Also, if rate (R) changes from R_1 to R_2 and time (T) changes from T_1 to T_2, but principal (P) is fixed, then change in SI $= \dfrac{P}{100}\times(R_1T_1 - R_2T_2)$.

Illustration 12: If simple interest on ₹600 increases by ₹30, when the rate % increases by 4% per annum, find out the time.

Solution: Here, $P = 600$, change in SI $= 30$, $R_1 - R_2 = 4$, $T = ?$

Using, change in SI $= \dfrac{PT}{100}\times(R_1 - R_2)$

we have, $30 = \frac{600T}{100} \times 4$

$\Rightarrow \quad T = \frac{5}{4}$, i.e., $1\frac{1}{4}$ years.

Illustration 13: If the simple interest on ₹1400 be more than the interest on ₹1000 by ₹60 in 5 years, find the out the rate per cent per annum.

Solution: Here, change in SI = 60, $P_1 - P_2 = 400$, $T = 5$, $R = ?$

Using change in SI $= \frac{RT}{100} \times (P_1 - P_2)$

We have, $60 = \frac{5R}{100} \times 400 \quad \Rightarrow \quad R = 3\%$

Illustration 14: If the simple interest on a certain sum at 4% per annum for 4 years is ₹80 more than the interest on the same sum for 3 years at 5% per annum, find out the sum.

Solution: Here, change in SI = 80, $R_1 = 4$, $R_2 = 5$, $T_1 = 4$, $T_2 = 3$, $P = ?$

Using change in SI $= \frac{P}{100} \times (R_1T_1 - R_2T_2)$

We have, $80 = \frac{P}{100} \times (4 \times 4 - 5 \times 3)$

$\Rightarrow \quad P =$ ₹8000.

08 If a debt of ₹Z is paid in 'n' number of instalments and if the value of each instalment is ₹a, then the borrowed (debt) amount is given by

$$Z = na + \frac{Ra}{100 \times b} \times \frac{n(n-1)}{2}$$

where, R = rate of interest per annum
b = number of instalments/year
$b = 1$, when each instalment is paid yearly
$b = 2$, when each instalment is paid half-yearly
$b = 4$, when each instalment is paid quarterly
$b = 12$, when each instalment is paid monthly.

Illustration 15: A sum of ₹2 is lent to be paid back in 3 equal monthly instalments of ₹1 each. Find the rate per cent.

Solution: Here, $Z =$ ₹2, $a =$ ₹1, $n = 3$, $b = 12$, $R = ?$

Using the formula

$$Z = na + \frac{Ra}{100 \times b} \times \frac{n(n-1)}{2},$$

we have, $2 = 3 \times 1 + \frac{R \times 1}{100 \times 12} \times \frac{3 \times 2}{2} \quad \Rightarrow \quad R = 400\%$

$\therefore$ The rate % p.a. is 400%

09 If a certain sum of money P lent out at SI amounts to A_1 in T_1 years and to A_2 in T_2 years, then

$$P = \frac{A_1T_2 - A_2T_1}{T_2 - T_1}$$

and, $$R = \frac{A_1 - A_2}{A_1T_2 - A_2T_1} \times 100\%$$

Illustration 16: If a certain sum of money at simple interest amounts to ₹5184 in 2 years and to ₹5832 in 3 years, what is the sum and the rate of interest?

Solution: Principal $= \frac{A_1T_2 - A_2T_1}{T_2 - T_1}$

$$\left[\begin{array}{l}\text{Here, } A_1 = 5184,\ A_2 = 5832\\ T_1 = 2,\ T_2 = 3\end{array}\right]$$

$$= \frac{5184 \times 3 - 5832 \times 2}{3 - 2} = ₹3888$$

and, Rate $= \frac{(A_2 - A_1) \times 100}{T_1A_2 - T_2A_1} = \frac{(5832 - 5184) \times 100}{2 \times 5832 - 3 \times 5184}$

$$= \frac{64800}{3888} = 16\frac{2}{3}\%$$

10 If a certain sum of money P lent out for a certain time T amounts to A_1 at R_1% per annum and to A_2 at R_2% per annum, then

$$P = \frac{A_2R_1 - A_1R_2}{R_1 - R_2}$$

and, $T = \frac{A_1 - A_2}{A_2R_1 - A_1R_2} \times 100$ years.

Illustration 17: A certain sum is invested for certain time. It amounts to ₹450 at 7% per annum. But, when invested at 5% per annum, it amounts to ₹350. Find out the sum and time.

Solution: Here, $A_1 = 450$, $R_1 = 7$, $A_2 = 350$, $R_2 = 5$.

Using the formula,

$$P = \frac{A_2R_1 - A_1R_2}{R_1 - R_2}$$

We get, $P = \frac{350 \times 7 - 450 \times 5}{7 - 5} =$ ₹100

Also, using the formula,

$$T = \left(\frac{A_1 - A_2}{A_2R_1 - A_1R_2}\right) \times 100$$

we get, $T = \left(\frac{450-350}{350\times 7-450\times 5}\right)\times 100 = 5$ years.

11 If an amount P_1 lent at simple interest rate of R_1% per annum, and another amount P_2 at simple interest rate of R_2% per annum, then the rate of interest for the whole sum is

$$R = \left(\frac{P_1R_1 + P_2R_2}{P_1 + P_2}\right).$$

Illustration 18: Mohan deposits ₹5000 in NSC at 2% per annum and ₹2000 in mutual funds at 4% per annum. Find out the rate of interest for the whole sum.

Solution: Here, $P_1 = 5000$, $R_1 = 2$, $P_2 = 2000$, $R_2 = 4$. Using the formula

$$R = \left(\frac{P_1R_1 + P_2R_2}{P_1 + P_2}\right)$$

We get, $R = \frac{5000\times 2 + 2000\times 4}{5000 + 2000} = 2\frac{4}{7}\%$

12 If a certain sum of money is lent out in n parts in such a manner that equal sum of money is obtained as simple interest on each part where interest rates are $R_1, R_2, ..., R_n$ and time periods are $T_1, T_2, ..., T_n$, respectively, then the ratio in which the sum will be divided in n parts is given by

$$\frac{1}{R_1T_1} : \frac{1}{R_2T_2} : ... \frac{1}{R_nT_n}.$$

Explanation

Let the n equal parts be $P_1, P_2, ..., P_n$ and, let I be the equal interest earned on each part.

Then, $P_1 = \frac{I\times 100}{R_1T_1}$

$P_2 = \frac{I\times 100}{R_2T_2}$

$\vdots \qquad\qquad \vdots$

$P_n = \frac{I\times 100}{R_nT_n}$.

$\therefore \quad P_1:P_2: ... :P_n = \frac{I\times 100}{R_1T_1} : \frac{I\times 100}{R_2T_2} : ... \frac{I\times 100}{R_nT_n}$

$= \frac{1}{R_1T_1} : \frac{1}{R_2T_2} : ... \frac{1}{R_nT_n}$.

Illustration 19: If a sum of ₹1600 is divided into two such parts that the simple interest on the first part for 2 $\frac{1}{2}$ years at the rate of 4% p.a. equals the simple interest on the second part for 5 years at the rate of 3% p.a., then find two such divisions of the sum.

Solution: Ratio of one part to other part of ₹1600

$$= \frac{1}{R_1T_1} : \frac{1}{R_2T_2}$$

$\therefore$ 1st part:2nd part $= \frac{1}{4\times\frac{5}{2}} : \frac{1}{3\times 5}$

[Here, R_1 = 4% p.a., $T_1 = \frac{5}{2}$ years, R_2 = 3% p.a., T_2 = 5 years]

or, 1st part:2nd part $= \frac{1}{10} : \frac{1}{15} = 3:2$

Sum of proportionals = 3 + 2 = 5

$\therefore$ 1st part $= \frac{3}{5}\times 1600 = ₹96$

and, 2nd part $= \frac{2}{5}\times 1600 = ₹640$.

13 When there is a change in principal (P), Rate (R) and Time (T), then the value of simple interest I also changes and is given by

$$\frac{I_1}{I_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

$$\Rightarrow \quad \frac{A_1 - P_1}{A_2 - P_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

as $I_1 = A_1 - P_1$ and $I_2 = A_2 - P_2$.

Illustration 20: If ₹85 amounts to ₹95 in 3 years, what ₹102 will amount to in 5 years at the same rate per cent?

Solution: Here, P_1 = ₹85, A_1 = ₹95, T_1 = 3 years, P_2 = ₹102, T_2 = 5 years, $R_1 = R_2 = R$ (say).

Then, using the formula

$$\frac{A_1 - P_1}{A_2 - P_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

We have, $\frac{95-85}{A_2 - 102} = \frac{85\times R\times 3}{102\times R\times 5}$

$\Rightarrow \quad A_2 - 102 = 20$

$\Rightarrow \quad A_2 = 122$

$\therefore$ The amount is ₹122.

14 Out of a certain sum P, $\frac{1}{a}$ part is invested at R_1%, $\frac{1}{b}$ part at R_2% and the remainder $\left(1-\frac{1}{a}-\frac{1}{b}\right)$ say

$\frac{1}{c}$ part at R_3%. If the annual income from all these investments is ₹A, then the original sum is given by

$$P = \left(\frac{A \times 100}{\frac{R_1}{a} + \frac{R_2}{b} + \frac{R_3}{c}} \right).$$

Illustration 21: Out of a certain sum, $\frac{1}{3}$ is invested at 3%, $\frac{1}{6}$ at 6% and the rest at 8%. If the annual income is ₹300, then the original sum is

Solution: Here,

$$\frac{1}{a} = \frac{1}{3}, \frac{1}{b} = \frac{1}{6},$$

$$\frac{1}{c} = 1 - \left(\frac{1}{3} + \frac{1}{6} \right) = \frac{1}{2},$$

R_1 = 3%, R_2 = 6%, R_3 = 8%,

A = ₹300.

$$\therefore \text{ The original sum } = \frac{A \times 100}{\frac{R_1}{a} + \frac{R_2}{b} + \frac{R_3}{c}}$$

$$= \frac{300 \times 100}{\frac{3}{3} + \frac{6}{6} + \frac{8}{2}} = \frac{30000}{6}$$

= ₹5000.

EXERCISE-I

1. The simple interest on ₹500 at 6% per annum from May 3rd to July 15th in the same year is:
(a) ₹9 (b) ₹6
(c) ₹4 (d) None of these

2. Mr Irani borrowed a sum of ₹10000 from a finance company for 6 years at 8% per annum. The amount returned by Mr Irani to the finance company is:
(a) ₹14800 (b) ₹12600
(c) ₹13300 (d) None of these

3. The principal that will yield ₹60 as simple interest at 6% per annum in 5 years is:
(a) ₹175 (b) ₹350
(c) ₹200 (d) None of these

4. The sum of money that will produce ₹1770 interest in $7\frac{1}{2}$ years at 8% simple interest per annum is:
(a) ₹2950 (b) ₹3120
(c) ₹2800 (d) None of these

5. If the simple interest on a certain sum of money after $6\frac{1}{4}$ years is 3/8 of the principal, then the rate of interest per annum is:
(a) 5% (b) 6%
(c) 4% (d) None of these

6. Rakesh borrowed ₹5000 from Ganesh at simple interest. If Ganesh received ₹500 more than his capital after 5 years, then the rate of interest per annum is:
(a) 2% (b) 3%
(c) 4% (d) None of these

7. The rate per cent per annum at which ₹1200 amount to ₹1440 in 4 years, is:
(a) 5% (b) 4%
(c) 6% (d) None of these

8. If simple interest on a certain sum of money is ₹256 and the rate of interest per annum equals the number of years, then the rate of interest is:
(a) 13% (b) 14%
(c) 16% (d) None of these

9. If the simple interest on a certain sum of money for 2 years is one-fifth of the sum, then the rate of interest per annum is:
(a) 9% (b) 10%
(c) 8% (d) None of these

10. If the simple interest on a certain sum of money is $\frac{4}{25}$ of the sum and the rate per cent equals the number of years, then the rate of interest per annum is:
(a) 2% (b) 3%
(c) 4% (d) None of these

11. If a certain sum of money borrowed at 5% per annum simple interest amounts to ₹1020 in 4 years, then the sum of money borrowed is:
(a) ₹850 (b) ₹925
(c) ₹750 (d) None of these

12. In what time ₹1200 will become ₹1344 at 6% per annum?

(a) $2\frac{1}{2}$ years (b) 3 years

(c) 2 years (d) None of these

13. In what time ₹8100 will produce the same income at 3% as ₹225 in 4 years at 3%.

(a) $\frac{1}{7}$ years (b) $\frac{1}{9}$ years

(c) $\frac{1}{6}$ years (d) None of these

14. If ₹1000 be invested at interest rate of 5% and the interest be added to the principal every 10 years, then the number of years in which it will amount to ₹2000 is:

(a) $16\frac{2}{3}$ years (b) $16\frac{1}{4}$ years

(c) 16 years (d) None of these

15. If ₹500 amounts to ₹725 at 9% simple interest in some time, what will ₹600 amount to at 11% in the same time?

(a) ₹870 (b) ₹930

(c) ₹910 (d) None of these

16. Sumit lends ₹10000 for 2 years at 20% per annum simple interest. After 1 year, he receives ₹6000. How much will he receive next year?

(a) ₹5900 (b) ₹6400

(c) ₹7200 (d) None of these

17. What principal will amount to ₹15000 at 10% per annum in 5 years?

(a) ₹10000 (b) ₹8700

(c) ₹10500 (d) None of these

18. The annual payment that will discharge a debt of ₹47250 due 3 years, hence at the rate of 5% simple interest is:

(a) ₹8000 (b) ₹10000

(c) ₹15000 (d) None of these

19. The annual instalment that will discharge a debt of ₹4200 due in 5 years at 10% simple interest is:

(a) ₹700 (b) ₹750

(c) ₹800 (d) None of these

20. If the amount obtained by Mahesh by investing ₹1500 for $2\frac{1}{2}$ years at the rate of 8% p.a. is equal to the amount obtained by Suresh by investing a certain sum for 2 years at 5% p.a. simple interest, then the sum invested by Suresh is:

(a) ₹$1636\frac{4}{11}$ (b) ₹1636

(c) ₹$1636\frac{1}{2}$ (d) None of these

21. A man invests ₹3965 in the names of his three daughters, Neeta, Sita and Gita in such a way that they would receive the same amount after 2, 3 and 4 years. If the rate of interest is 5% p.a., then the amount invested for Neeta, Sita and Gita is:

(a) ₹1380, ₹1320, ₹1265
(b) ₹1330, ₹1360, ₹1380
(c) ₹1265, ₹1320, ₹1340
(d) None of these

22. A sum of money at simple interest becomes four times in 24 years. The rate per cent of interest per annum is:

(a) $13\frac{3}{4}\%$ (b) $12\frac{1}{2}\%$

(c) $11\frac{3}{4}\%$ (d) None of these

23. In how many years will a sum of money treble itself at 10% per annum simple interest?

(a) 15 years (b) 19 years

(c) 20 years (d) None of these

24. A sum of money doubles itself in 8 years. In how many years will it treble?

(a) 16 years (b) 15 years

(c) 14 years (d) None of these

25. A sum was put at simple interest at a certain rate for 4 years. Had it been put at 2% higher rate, it would have fetched ₹56 more. Find the sum.

(a) ₹680 (b) ₹700

(c) ₹720 (d) None of these

26. If the interest on ₹800 be more than the interest on ₹400 by ₹40 in 2 years, then the rate of interest per annum is:

(a) 5% (b) $5\frac{1}{2}\%$

(c) 6% (d) None of these

27. If the difference between the simple interest on a certain sum for 4 years at $2\frac{1}{2}\%$ per annum and the simple interest on the same sum for the same period at 3% per annum is ₹60, then the sum is:

(a) ₹3000 (b) ₹2900

(c) ₹3100 (d) None of these

28. If a certain sum of money at simple interest amounts to ₹2800 in 2 years and ₹3250 in 5 years, then the rate of interest per annum is:

(a) 4% (b) 6%
(c) 5% (d) None of these

29. If a certain sum of money amounts to ₹1760 in two years and ₹2000 in 5 years at simple interest, then the sum is:

(a) ₹1960 (b) ₹1590
(c) ₹1600 (d) None of these

30. A certain sum is invested for certain time. It amounts to ₹450 at 7% per annum. But when invested at 5% per annum, it amounts to ₹350. Find the sum.

(a) ₹60 (b) ₹100
(c) ₹120 (d) None of these

31. A certain sum is invested for T years. It amounts to ₹400 at 10% per annum. But, when invested at 4% per annum, it amounts to ₹200. Find the time (T).

(a) 41 years (b) 39 years
(c) 50 years (d) None of these

32. If a sum of ₹9 is lent to be paid back in 10 equal monthly instalments of ₹1 each, then the rate of interest is:

(a) $266\frac{2}{3}\%$ (b) $265\frac{3}{4}\%$
(c) 266% (d) None of these

33. A sum of ₹7700 is to be divided among three brothers Vikas, Vijay and Viraj in such a way that simple interest on each part at 5% per annum after 1, 2 and 3 years, remains equal. The share of Vikas is more than that of Viraj by:

(a) ₹2800 (b) ₹2500
(c) ₹3000 (d) None of these

34. If simple interest on a certain sum of money for 4 years at 5% p.a. is same as the simple interest on ₹560 for 10 years at the rate of 4% p.a., then the sum of money is:

(a) ₹1190 (b) ₹1120
(c) ₹1210 (d) None of these

35. Mr Mani invested an amount of ₹12000 at a simple interest rate of 10% per annum and another amount at a simple interest rate of 20% per annum. The total interest earned at the end of one year on the total amount invested became 14% per annum. Find the total amount invested.

(a) ₹20000 (b) ₹20800
(c) ₹21000 (d) None of these

36. Mr Gupta deposits ₹3000 in a bank at 10% per annum and ₹5000 in another bank at 8% per annum. The rate of interest for the whole sum is:

(a) $8\frac{1}{2}\%$ (b) $8\frac{3}{4}\%$
(c) 8% (d) None of these

37. A person invested $\frac{2}{3}$ of his capital at 3%, $\frac{1}{6}$ at 6% and the remainder at 12%. If his annual income is ₹25, then the capital is:

(a) ₹490 (b) ₹510
(c) ₹500 (d) None of these

38. The simple interest on a sum of money will be ₹600 after 10 years. If the principal is trebled after 5 years, then what will be the total interest at the end of the tenth year?

(a) ₹1200 (b) ₹1190
(c) ₹1210 (d) None of these

39. ₹1500 is invested at a rate of 10% simple interest and interest is added to the principal after every 5 years. In how many years will it amount to ₹2500.

(a) $6\frac{1}{9}$ years (b) $6\frac{1}{4}$ years
(c) 7 years (d) None of these

40. Sumit lent some money to Mohit at 5% per annum simple interest. Mohit lent the entire amount to Birju on the same day at $8\frac{1}{2}\%$ per annum. In this transaction, after a year Mohit earned a profit of ₹350. Find out the sum of money lent by Sumit to Mohit.

(a) ₹9000 (b) ₹10000
(c) ₹10200 (d) None of these

41. Brinda borrowed ₹1000 to build a hut. She pays 5% simple interest. She lets the hut to Ramu and receives a rent of $₹12\frac{1}{2}$ per month from Ramu. In how many years Brinda would clear off the debt?

(a) 10 years (b) $10\frac{1}{4}$ years
(c) $10\frac{1}{2}$ years (d) None of these

42. The rate of interest on a sum of money is 4% per annum for the first 2 years, 6% per annum for the next 4 years, and 8% per annum for the period beyond 6 years. If the simple interest accrued by the sum for a total period of 9 years is ₹1120, then the sum is:

(a) ₹2400 (b) ₹2200
(c) ₹2000 (d) None of these

EXERCISE-2
(BASED ON MEMORY)

1. The simple interest on a sum for 8 years is ₹47500. The rate of 10% per annum and for the next 3 years is 15% per annum. What is the value (in ₹) of sum?

(a) 50000 (b) 60000
(c) 45000 (d) 62500

[SSC CHSL (10+2) Tier-I CBE, 2018]

2. A sum of ₹8000 is divided into two parts. The simple interest on first part at the rate of 21% per annum is equal to the simple interest on second part at the rate of 35% per annum. What is the interest (in ₹) of each part?

(a) 1050 (b) 840
(c) 1400 (d) 1220

[SSC CHSL (10+2) Tier-I CBE, 2018]

3. An investor invested his saving in the stock market. The value of his investments increased by 12% and 9% in the first year and the second year respectively. If the value of his investments after two years became ₹97,664 then how much had he invested (in ₹)?

(a) 81000 (b) 75000
(c) 80000 (d) 72000

[SSC CGL Tier-II CBE, 2018]

4. In how much time (in years) will a sum of ₹9500 amounts to ₹17622.5 at the rate of 4.5% per annum at simple interest?

(a) 21 (b) 23
(c) 19 (d) 20

[SSC Delhi Police Constable, 2017]

5. Dalajit lent ₹10800 to Jaabir for 3 years and ₹7500 to kabir for 2 years on simple interest and received ₹1422 in all from both of them as interest. The rate of interest per annum is

(a) 3.5 per cent (b) 4 per cent
(c) 3 per cent (d) 4.5 per cent

[SSC CGL Tier-I CBE, 2017]

6. Deepak lent ₹8800 to Jaichand for 13 years on simple interest at the same rate of interest and received ₹14432 in all from both of them as interest. The rate of interest per annum is

(a) 8.5 per cent (b) 9 per cent
(c) 9.5 per cent (d) 8 per cent

[SSC CGL Tier-I CBE, 2017]

7. A sum was doubled with $12\frac{1}{2}\%$ rate of simple interest, per annum. Then taken for that sum is:

(a) $8\frac{1}{2}$ years (b) 8 years
(c) 10 years (d) $12\frac{1}{2}$ years

[SSC Matric Level MTS, 2017]

8. What is the simple interest on ₹5400 in 5 years at the rate of 12% per annum?

(a) ₹2700 (b) ₹2950
(c) ₹3120 (d) ₹3240

[SSC CAPFs ASI and Delhi Police SI, 2017]

9. A certain sum becomes ₹1020 in 5 years and ₹1200 in B years at simple interest. What is the value of Principal?

(a) ₹820 (b) ₹780
(c) ₹700 (d) ₹720

[SSC CAPFs ASI and Delhi Police SI, 2017]

10. A certain sum becomes 5 times in 3 years, at simple interest, then in how many years it will become 13 times?

(a) 6 (b) 15
(c) 9 (d) 12

[SSC CAPFs ASI and Delhi Police SI, 2017]

11. A certain sum of money amounts to ₹918 in 2 years and ₹969 in 3.5 years at simple interest, What is the rat of intrest (in % per annum)?

(a) 4 (b) 5
(c) 6 (d) 8

[SSC CGL Tier-I CBE, 2017]

12. Simple interest received by a person in 10 years on a principal of ₹9500 is 130% of the principal. What is the rate of interest (in %) per annum?

(a) 12 (b) 13
(c) 15 (d) 19

[SSC CGL Tier-I CBE, 2017]

13. An amount fetched a total simple interest of ₹3200 at the rate of 6.25% per year in 4 years. What is the amount (in ₹)?

(a) 13800 (b) 11800
(c) 12800 (d) 14800

[SSC CGL Tier-I CBE, 2017]

14. The effective annual rate of interest corresponding to a nominal rate of 13% per annum payable half-yearly is

(a) 26% (b) 26.85%
(c) 13% (d) 13.42%

[SSC CHSL (10+2) Tier-I CBE, 2017]

15. A person lent ₹10000 to B for 3 years and ₹6000 to C for 4 years on simple interest at same rate of interest and received ₹5400 in all from both of them as interest. What is the rate of the interest (in%)?

(a) 10 (b) 12.5
(c) 15 (d) 20

[SSC CHSL (10+2) Tier-I CBE, 2017]

16. Albert invested and amount of x rupees in a fixed deposit scheme offering 10% per annum for 1st Year and 15% per annum for 2nd year and received an amount of ₹20,240 after two years. What is x (in ₹)?

(a) 15000 (b) 16000
(c) 14000 (d) 18000

[SSC CGL Tier-I CBE, 2017]

17. Alipta got some amount of money from her father. In how many years will the ratio of the money and the interest obtained from it be 10:3 at the rate of 6% simple interest per annum?

(a) 7 years (b) 3 years
(c) 5 years (d) 4 years

[SSC CGL Tier-I (CBE), 2016]

18. The simple interest on a certain sum of money at the rate of 5% per annum for 8 years is ₹840. Rate of interest for which the same amount of interest can be received on the same sum, after 5 years is:

(a) 7% per annum (b) 8% per annum
(c) 9% per annum (d) 10% per annum

[SSC CHSL (10+2) Tier-I (CBE), 2016]

19. A sum of ₹2800 is divided into two parts in such a way that the interest on both the parts is equal. If the first part is lent at 9% p.a. for 5 years and second part is for 6 years at 10% 9.a., find the two sums.

(a) ₹1800, ₹1000 (b) ₹1600, ₹1200
(c) ₹1400, ₹1400 (d) ₹1300, ₹1500

[SSC CAPFs (CPO) SI and ASI, Delhi Police, 2016]

20. If the simple interest of ₹1 for month is paisa, then the rate per cent per annum will be

(a) 10% (b) 8%
(c) 12% (d) 6%

[SSC CGL Tier-I (CBE), 2016]

21. In simple interest rate per annum a certain sum amounts to ₹5,182 in 2 years and ₹5,832 in 3 years. The principal rupees is

(a) ₹2882 (b) ₹5000
(c) ₹3882 (d) ₹4000

[SSC CGL Tier-I (CBE), 2016]

22. For what sum will the simple interest at R% per annum for 2 years will be R?

(a) ₹$\frac{100}{2R}$ (b) ₹50

(c) ₹$\frac{100}{R}$ (d) ₹$\frac{200}{R}$

[SSC CGL Tier-I (CBE), 2016]

23. A certain sum doubles in 7 years at simple interest. The same sum under the same interest rate will become 4 times in how many years.

(a) 14 (b) 28
(c) 21 (d) 10

[SSC CPO SI, ASI, 2016]

24. A certain sum of money amounts to ₹2200 at 5% p.a. rate of interest, ₹2320 at 8% interest in the same period of time, The period of time is:

(a) 3 years (b) 4 years
(c) 5 years (d) 2 years

[SSC CAPFs (CPO) SI and ASI, Delhi Police, 2016]

25. At what per cent of simple interest will a sum of money double itself in 15 years?

(1) $6\frac{1}{3}\%$ (2) $6\frac{2}{3}\%$

(3) $6\frac{1}{2}\%$ (4) 6%

[SSC CGL Tier-I (CBE), 2016]

26. The rate of simple interest for which a sum money becomes 5 times of itself in 8 years is:

(a) 30% (b) 40%
(c) 50% (d) 55%

[SSC CGL Tier-I (CBE), 2016]

27. If a borrowed ₹P at x% and B borrowed ₹Q (> P) at y% per annum at simple interest at the same time, then the amount of their debts will be equal after

(a) $100\left(\frac{Q-P}{Px-Qy}\right)$ years

(b) $100\left(\frac{Px-Qy}{Q-P}\right)$ years

(c) $100\left(\frac{Px-Qy}{P-Q}\right)$ years

(d) $100\left(\frac{P-Q}{Px-Qy}\right)$ years

[SSC CGL Tier-II, 2016]

28. A money lender claims to lend money at the rate of 10% per annum simple interest, However, he takes the interest in advance when he lends a sum for one year. At what interest rate does he lend the money actually?

(a) 10% (b) $10\frac{1}{9}\%$

(c) 11% (d) $11\frac{1}{9}\%$

[SSC CPO SI, ASI, 2016]

29. A man buys a watch for ₹1950 in cash and sells it for ₹2200 at a credit of 1 year. If the rate of interest be 10% per annum, then how much profit or loss will he have?

(a) ₹55 gain (b) ₹50 Profit
(b) ₹30 gain (d) ₹30 Profit

[SSC CGL Tier-I (CBE), 2016]

30. A money lender lends ₹400 for 3 years to a person and lends ₹500 for 4 years to the other person at the same rate of simple interest. If altogether he receives ₹160 as interest per annum?

(a) 5% (b) 7%
(c) 9% (d) 10%

[SSC CGL Tier-I (CBE), 2016]

31. The simple interest on a sum of money is $\frac{8}{25}$ of the sum. If the number of years is numerically half the rate percent per annum, then the rate percent per annum is

(a) 4 (b) $6\frac{1}{4}$

(c) 5 (d) 8

[SSC, 2015]

32. A sum of money lent out at simple interest amounts to ₹720 after 2 years and ₹1020 after a further period of 5 years. Find the principal.

(a) ₹600 (b) ₹1740
(c) ₹120 (d) ₹6000

[SSC, 2015]

33. Ram deposited a certain sum of money in a company at 12% per annum simple interest for 4 years and deposited equal amount in fixed deposit in a bank for 5 years at 15% per annum simple interest. If the difference in the interest from two sources is ₹1350 then the sum deposited in each case is:-

(a) ₹5000 (b) ₹4000
(c) ₹6500 (d) ₹3000

[SSC, 2015]

34. Prakash lends a part of ₹20,000 at 8% simple interest and remaining at $\frac{4}{3}\%$ simple interest. His total income after a year was ₹800. Find the sum lent at 8%.

(a) ₹8,000 (b) ₹12,000
(c) ₹6,000 (d) ₹10,000

[SSC, 2014]

35. Nitin borrowed some money at the rate of 6% p.a. for the first three years, 9% p.a. for the next five years and 13% p.a. for the period beyond eight years. If the total interest paid by him at the end of eleven years is ₹8,160, the money borrowed by him was:

(a) 12,000 (b) 6,000
(c) 8,000 (d) 10,000

[SSC Assistant Grade III, 2013]

36. A sum of ₹16800 is divided into two parts. One part is lent at a simple rate of interest 6% per annum and the other at 8% per annum. After 2 years the total sum received is ₹19000. The sum lent at the rate of 6% simple interest is:

(a) ₹12200 (b) ₹12000
(c) ₹11000 (d) ₹10000
(e) None of these

[IBPS PO/MT, 2013]

37. Arun invested a sum of money at a certain rate of simple interest for a period of four years. Had he invested the same sum for a period of six years, the total interest earned by him would have been fifty per cent more than the earlier interest amount? What was the rate of interest per cent per annum?

(a) 4 (b) 8
(c) 5 (d) Cannot be determined

[Gramin Bank U.P. (SO) Examination, 2012]

38. The simple interest on a sum of money is $\frac{1}{9}$th of the principal and the number of years is equal to the rate per cent per annum. The rate per cent per annum is equal to:

(a) 3% (b) $\frac{1}{3}\%$

(c) $\frac{1}{10}\%$ (d) $3\frac{1}{3}\%$

[SSC Assistant Grade III, 2012]

39. Arun lends ₹20,000 to two of his friends. He gives ₹12,000 to the first at 8% p.a. simple interest. Arun wants to makes a profit of 10% on the whole. The simple interest rate at which he should lend the remaining sum of money to the second friend is:

(a) 8% (b) 16%
(c) 12% (d) 13%

[SSC, 2012]

40. In what time will a sum of money double itself at the rate of 20% per annum simple interest

(a) 10 years (b) 5 years
(c) 2 years (d) 14 years

[SSC (GL), 2011]

41. 800 becomes ₹956 in 3 years at a certain rate of simple interest. If the rate of interest is increased by 4%, what amount will ₹800 become in 3 years?

(a) ₹1020.80 (b) ₹1025
(c) ₹1052 (d) ₹1050

[SSC (GL), 2011]

42. Simple interest on a certain sum is $\frac{16}{25}$ of the sum. The rate per cent if the rate per cent and time (in years) are equal, is:

(a) 6% (b) 8%
(c) 10% (d) 12%

[SSC (GL), 2011]

43. If the simple interest on ₹x at a rate of a% for m years is same as that on ₹y at a rate of a^2% for m^2 years, then x:y is equal to:

(a) m:a (b) am:1

(c) $\frac{1}{m}:\frac{1}{a}$ (d) $\frac{1}{am}:1$

[SSC, 2011]

44. A took two loans altogether of ₹1200 from B and C. B claimed 14% simple interest per annum, while C claimed 15% per annum. The total interest paid by A in one year was ₹172. Then a borrowed:

(a) ₹800 from C (b) ₹625 from C
(c) ₹400 from B (d) ₹800 from B

[SSC, 2011]

45. A person has left an amount of ₹1,20,000 to be divided between his two sons aged 14 years and 12 years such that they get equal amounts when each attains 18 years of age. If the amount gets a simple interest of 5% per annum, the younger son's share at present is:

(a) ₹48,800 (b) ₹57,600
(c) ₹62,400 (d) ₹84,400

[SSC, 2011]

46. The simple interest accrued on an amount of ₹22,500 at the end of four years is ₹10,800. What would be the compound interest accrued on the same amount at the same rate of interest at the end of two years?

(a) ₹16,908 (b) ₹5,724
(c) ₹28,224 (d) ₹8,586
(e) None of these

[IBPS PO/MT, 2011]

47. The simple interest accrued on a sum of certain principal is ₹1200 in 4 years at the rate of 8% per annum.. What would be the simple interest accrued on thrice of that principal at the rate of 6% per annum in 3 years?

(a) ₹2025 (b) ₹3025
(c) ₹2250 (d) ₹2150

[OBC PO, 2010]

48. A man invested $\frac{1}{3}$ of his capital at 7%, $\frac{1}{4}$ at 8% and the remaining at 10% rate of simple interest. If his annual income from interests is ₹561, then the capital invested was:

(a) ₹6000 (b) ₹5600
(c) ₹6600 (d) ₹7200

[SSC, 2010]

49. A person rece a simple interest of ₹1,000 on a certain principal at the rate of 5% p.a. in 4 years. What compound interest will would that person receive on twice the principal in two years at the same rate?

(a) ₹1,000 (b) ₹1,005
(c) ₹11,025 (d) ₹10,125
(e) None of these

[Punjab and Sind Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (b)	**2.** (a)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (c)	**9.** (b)	**10.** (c)	**11.** (a)	**12.** (c)	**13.** (b)
14. (a)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (a)	**21.** (a)	**22.** (b)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (a)
27. (a)	**28.** (b)	**29.** (c)	**30.** (b)	**31.** (c)	**32.** (a)	**33.** (a)	**34.** (b)	**35.** (a)	**36.** (b)	**37.** (c)	**38.** (a)	**39.** (a)
40. (b)	**41.** (a)	**42.** (c)										

EXERCISE-2

1. (a)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (d)	**10.** (c)	**11.** (a)	**12.** (b)	**13.** (c)
14. (d)	**15.** (a)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)	**21.** (c)	**22.** (b)	**23.** (c)	**24.** (d)	**25.** (b)	**26.** (c)
27. (a)	**28.** (d)	**29.** (b)	**30.** (a)	**31.** (d)	**32.** (a)	**33.** (a)	**34.** (a)	**35.** (c)	**36.** (a)	**37.** (d)	**38.** (d)	**39.** (d)
40. (b)	**41.** (c)	**42.** (b)	**43.** (b)	**44.** (d)	**45.** (b)	**46.** (b)	**47.** (a)	**48.** (c)	**49.** (e)			

EXPLANATORY ANSWERS

EXERCISE-1

1. (b) Time from May 3rd to July 15th

= 28 days of May + 30 days of June and 15 days of July

= 73 days = $\frac{73}{365}$ years, i.e., $\frac{1}{5}$ years.

$\therefore\ I = \frac{P \times R \times T}{100} = \frac{500 \times 6 \times \frac{1}{5}}{100} = ₹6.$

2. (a) We have, P = ₹10000, R = 8% per annum, T = 6 years.

$\therefore\ I = \frac{P \times R \times T}{100} = \frac{10000 \times 8 \times 6}{100} = ₹4800$

$\therefore\ A = P + I = 10000 + 4800 = ₹14800$

Thus, Mr Irani returned ₹14800 to the finance company.

3. (c) Here, I = ₹60, R = 6% per annum, T = 5 years.

$\therefore$ Principal $(P) = \frac{100 \times I}{R \times T} = \frac{100 \times 60}{6 \times 5} = ₹200.$

4. (a) Here, I = ₹1770, R = 8% per annum, $T = \frac{15}{2}$ years.

$\therefore$ Principal $(P) = \frac{100 \times I}{R \times T} = \frac{100 \times 1770}{8 \times \frac{15}{2}}$

= ₹2950.

5. (b) Let the sum of money be ₹x.

Then, simple interest = $\frac{3}{8}x$

Also, time = $6\frac{1}{4}$ years, i.e., $\frac{25}{4}$ years

$\therefore$ Rate $(R) = \frac{100 \times I}{P \times T} = \frac{100 \times \frac{3x}{8}}{x \times \frac{25}{4}}$

$= \frac{100 \times 3}{2 \times 25} = 6\%.$

6. (a) Here, P = ₹5000, I = ₹500, T = 5 years.

Therefore, using the formula

$R = \frac{100 \times I}{P \times T}$

We have, rate of interest $(R) = \frac{100 \times 500}{5000 \times 5} = 2\%$ p.a.

7. (a) We have, P = ₹1200, T = 4 years,

I = 1440 −1200 = ₹240.

$\therefore$ Rate $(R) = \frac{100 \times I}{P \times T} = \frac{100 \times 240}{1200 \times 4} = 5\%$ per annum.